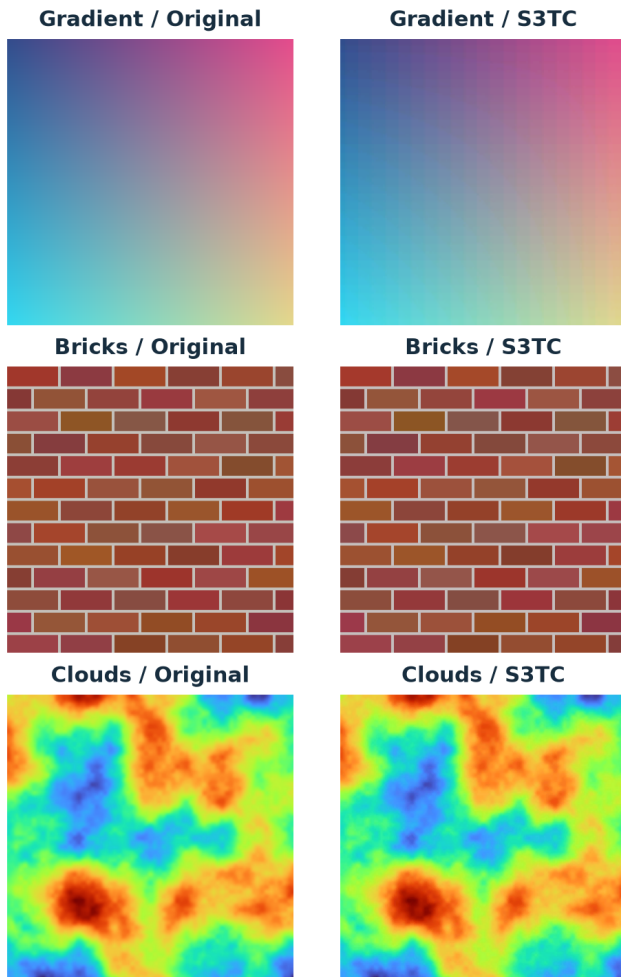


S3TC baseline

Code: <https://github.com/olliearrison/Neural-Graphics-A1.git>

All textures are 512 x 512 RGB: 768 KiB raw. 1 KiB = 1,024 bytes. Ratios are compressed/raw. PSNR uses full-image normalized RGB MSE before PNG rounding.



Texture	PSNR (dB)	Ratio
Gradient	43.93	16.67%
Bricks	42.02	16.67%
Clouds	42.87	16.67%

Each S3TC texture uses 128 KiB (6.00x smaller than raw RGB).

Endpoint selection

For each 4 x 4 block, try principal-axis extremes and componentwise color bounds. Quantize to RGB565, assign nearest palette entries, and refine endpoints by least squares for eight iterations, retaining the lowest-error pair.

Visible limitations

Gradient: blockwise color steps interrupt the smooth ramp. Bricks: slight shade shifts and edge-color errors; most blocks contain few colors, so damage is mild. Clouds: small block and color steps appear in smooth regions. Each block has only four colors on one RGB line.

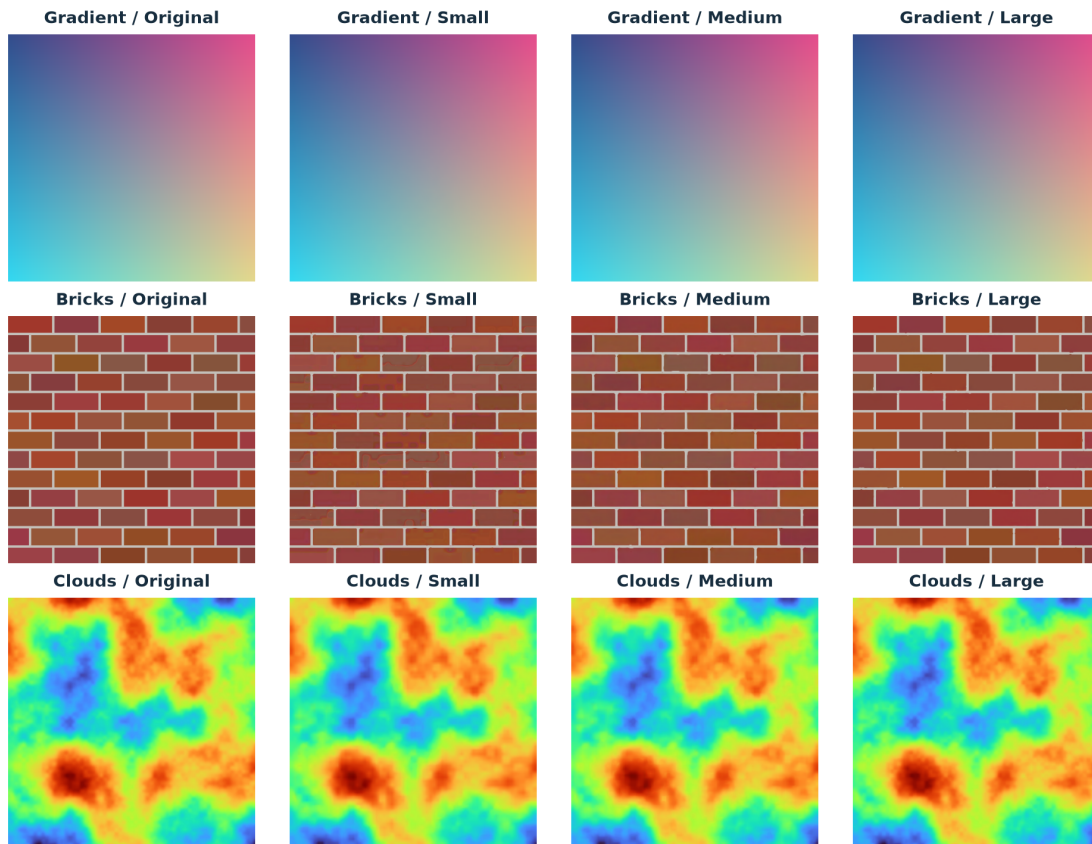
Why the extra green bit?

Human brightness perception is most sensitive to green, so its extra precision reduces visible luminance errors.

RGB565 approximation

The permitted q/31 or q/63 decode closely approximates integer bit expansion. For example, 5-bit q=3 gives 3/31 versus 24/255 after bit replication; finite-byte rounding causes the small difference.

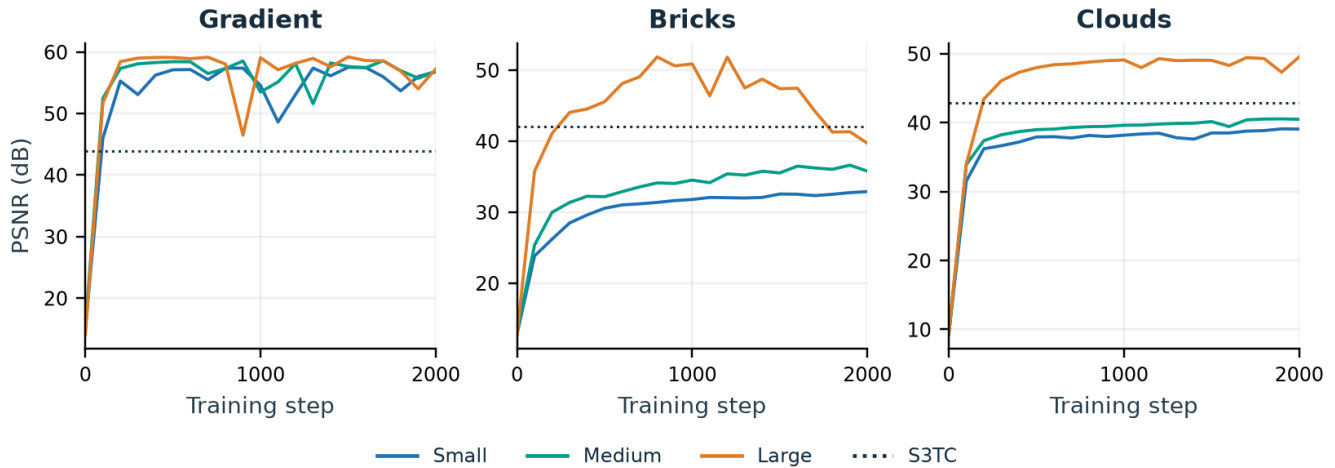
Neural reconstructions and results



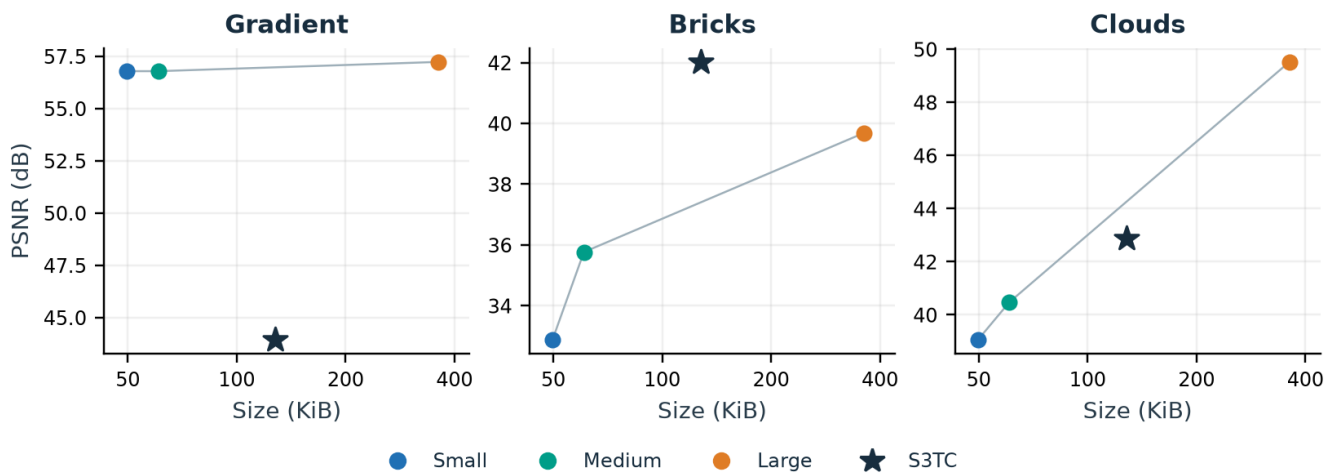
Float32 results after 2,000 steps. Size = all grid and MLP parameters x 4 bytes; raw RGB is 768 KiB. File headers are excluded.

Texture / model	PSNR (dB)	Size (KiB)	Ratio
Gradient / Small	56.79	49.76	6.48%
Gradient / Medium	56.79	60.76	7.91%
Gradient / Large	57.23	361.26	47.04%
Bricks / Small	32.88	49.76	6.48%
Bricks / Medium	35.76	60.76	7.91%
Bricks / Large	39.68	361.26	47.04%
Clouds / Small	39.06	49.76	6.48%
Clouds / Medium	40.47	60.76	7.91%
Clouds / Large	49.52	361.26	47.04%

Training and size-quality comparison



Full-image PSNR every 100 steps for all nine fits; dotted lines show S3TC.



Measured float32 size versus PSNR; stars show S3TC. Connecting lines only guide the eye.

Gradient: Large is best at 57.23 dB, but adds only 0.44 dB over Small: the smooth signal needs little capacity. All three exceed S3TC quality at the final step.

Bricks: Large is best at 39.68 dB because its finer grid represents narrow mortar edges better. Medium gains 2.89 dB over Small; all final neural fits trail S3TC (42.02 dB).

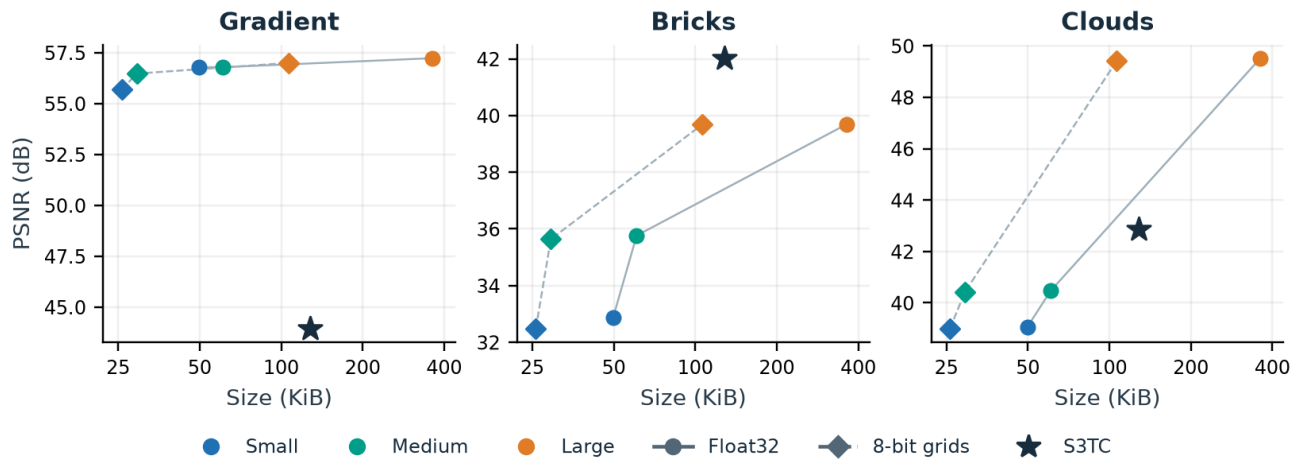
Clouds: Large is best at 49.52 dB: its finer grid and extra features capture detail across scales. Medium gains 1.41 dB over Small. Only Large exceeds S3TC quality (42.87 dB).

Size comparison for each texture: Small (6.48%) and Medium (7.91%) beat S3TC's 16.67% ratio; Large float32 (47.04%) does not.

Eight-bit grid quantization

Each grid uses its own uniform 8-bit quantizer; MLP parameters remain float32. Sizes include the uint8 indices and one float32 (lo, scale) pair per grid.

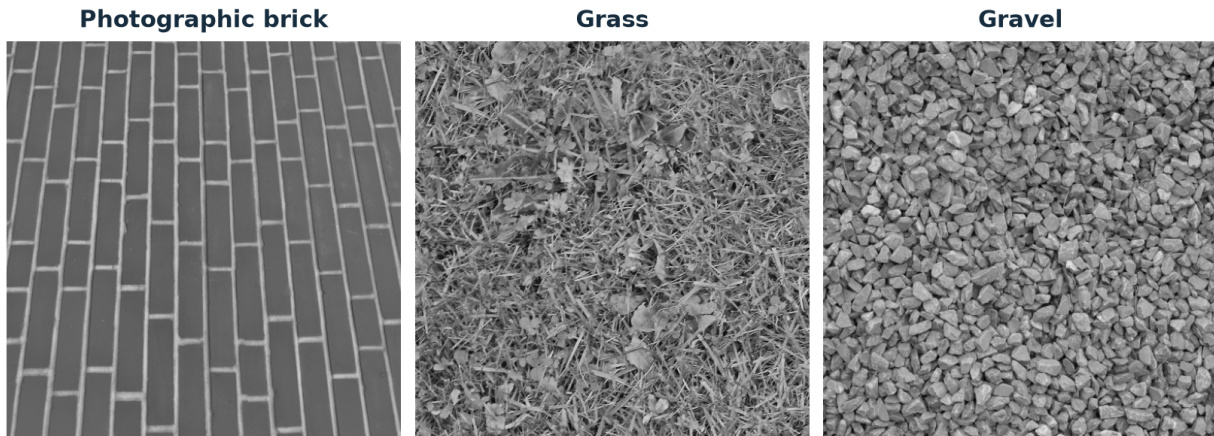
Texture / model	Before dB	After dB	Loss dB	KiB	Ratio	Factor
Gradient / Small	56.786	55.681	+1.105	25.77	3.36%	29.80x
Gradient / Medium	56.786	56.471	+0.314	29.29	3.81%	26.22x
Gradient / Large	57.228	56.994	+0.234	106.29	13.84%	7.23x
Bricks / Small	32.877	32.476	+0.401	25.77	3.36%	29.80x
Bricks / Medium	35.765	35.636	+0.129	29.29	3.81%	26.22x
Bricks / Large	39.684	39.689	-0.004	106.29	13.84%	7.23x
Clouds / Small	39.059	38.993	+0.065	25.77	3.36%	29.80x
Clouds / Medium	40.473	40.426	+0.047	29.29	3.81%	26.22x
Clouds / Large	49.521	49.407	+0.114	106.29	13.84%	7.23x



The P6 size-quality plot with quantized grids added as diamonds. All quantized models are smaller than S3TC. Quantized Large also beats S3TC PSNR on gradient and clouds.

Rounding perturbs the features supplied to the decoder. The largest loss is 1.105 dB for Small on the gradient, whose original error is already tiny; other losses are at most 0.401 dB. Large bricks improves by 0.004 dB because rounding happens to reduce residual error. The MLP staying float32 makes total savings smaller than 4x.

Three additional textures



CC0 samples from [scikit-image v0.25.2](#): CC0Textures Bricks25, linolafett's Grass 01, and CC0Textures Gravel04. The 512 x 512 grayscale sources are repeated into RGB without resizing.

Texture	Small	Medium	Large
Stored size (KiB)	49.76	60.76	361.26
Photographic brick	32.26 dB	35.12 dB	43.38 dB
Grass	19.39 dB	19.88 dB	25.89 dB
Gravel	21.46 dB	22.64 dB	31.04 dB

Each image is a fresh fit under all three architectures. Sizes count float32 grid and MLP parameters; the raw RGB reference is 768 KiB.

Photographic brick: regular mortar boundaries and smooth faces are relatively easy to represent. Large restores sharp boundaries and reaches 43.38 dB; the smaller models trade edge precision for smaller storage.

Grass: irregular, overlapping fine blades carry high-frequency detail. Small and Medium blur this structure; Large improves to 25.89 dB but still loses fine strands. It has the lowest quality at every model size.

Gravel: stones form larger coherent shapes than grass strands, so quality is higher at the same sizes. Large recovers much more boundary detail at 31.04 dB; Small and Medium remain compact but visibly smoother.

AI disclosure

OpenAI Codex developed the implementation, tests and report from an initial sampler draft; sourced the additional textures; ran training and verification; and drafted the figures and analysis. Reported metrics were computed by the executed code.